

Gaurav Tiwari

Boston, MA

Research Data Scientist



Lead investigator specializing in the integration of high-dimensional multi-omic data and advanced statistical genetics. Recognized for developing scalable computational models that bridge the gap between genomic discovery and clinical application. Extensive record of leadership in international research consortia and a primary contributor to flagship journals of the American Heart Association (AHA), focused on driving the next generation of cardiovascular risk prediction and patient care.



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EDUCATION

Master of Science (M.S.) - Advanced Data Analytics

Sep 2021 - Dec 2022

Boston University, Boston, MA

GPA: 3.93/4

Master of Business Administration (M.B.A.) - Finance

Jul 2016 - Apr 2018

Banaras Hindu University, Varanasi, India

GPA: 7.96/10

Bachelor of Technology (B.Tech.) - Electronics Engineering

Apr 2010 - Apr 2014

VIT University, Vellore, India

GPA: 8.04/10

WORK EXPERIENCE

Senior Data Analyst – Cardiovascular Research

Jan 2023 – Present

Beth Israel Deaconess Medical Center, Boston, MA

- Pre-processed, managed and analyzed large genomic, proteomics and metabolomics databases from epidemiological and clinical studies using R and Python in cardiovascular research data.
- Designed and executed several advanced analytical and ML models and led the team in predictive analysis of incident events and other important response variables.
- Visualized the important results using R, Python, Spark to provide insights in the data and results.
- Familiarity and established working knowledge of tools to investigate cardiovascular metabolomics with publicly available data sources.
- Spearheaded original research initiatives investigating the genetic architecture of cardiovascular and cardiometabolic diseases; work has been recognized and published in premier, high-impact journals including Nature and Circulation.
- Engineered end-to-end computational frameworks and scalable analytical pipelines for high-dimensional biomedical data, integrating advanced feature engineering and automated model validation to accelerate discovery cycles.
- Developed sophisticated machine learning architectures, employing both supervised and unsupervised learning to pioneer novel risk prediction models and diagnostic biomarker discovery with clinical utility.
- Directed comprehensive statistical genetics studies, including high-resolution Genome-Wide Association Studies (GWAS) and Phenome-Wide Association Studies (PheWAS) to identify causal variants, contributing to the global understanding of heart disease.
- Established and validated Polygenic Risk Score (PRS) methodologies optimized for diverse populations, translating genetic risk into clinical insights for diverse populations as featured in Journal of the American College of Cardiology (JACC).
- Orchestrated integrative multi-omic analyses, bridging proteomics and metabolomics with longitudinal clinical phenotypes to elucidate the molecular mechanisms underlying disease outcomes.
- Utilized network-based systems biology and functional enrichment methodologies to map biological pathways, providing mechanistic evidence for findings published in American Heart Association (AHA) flagship journals.

- Harnessed massive biobank-scale datasets (e.g., UK Biobank, All of Us) and EHR-linked longitudinal cohorts, implementing rigorous cross-cohort replication to ensure the global validity and reproducibility of research findings.
- Established gold-standard computational protocols, maintaining rigorous data quality control and reproducible research workflows to ensure the integrity of high-impact scientific outputs.
- Fostered cross-disciplinary synergy by leading collaborations between cardiologists, epidemiologists, and translational scientists, while providing specialized methodological mentorship to doctoral and post-doctoral researchers.
- Drove institutional growth through substantial contributions to federal grant applications (e.g., NIH R01/U01), consortium-level research strategies, and the long-term development of cardiovascular research programs.

Data Scientist Intern

May 2022 - Aug 2022

Vertex Pharmaceuticals Inc, (Atrium Payroll Services), Boston, MA

- Deep dive into Pre-Clinical Data generated by Lab and FDA and did Data cleaning, Data Preprocessing, and relevant Data Extraction from database, SharePoint, Web mining using Python and SQL.
- Created a consolidated database of all Lab studies which reduced execution time to 50 %.
- Designed ML models and other statistical techniques to forecast key business drivers and predict drug safety parameters to reduce time for pre-clinical safety assessments.
- Designed and automated pipeline and supported end-to-end for ML models transitioning to production from data engineering, validation and implementation and performance maintenance and monitoring.
- Implemented Dashboards on Power Bi and Tableau Desktop to answer key business problems and help in data visualization and to provide insights and metrics.

Graduate Teaching Assistant/Research Assistant

Jan 2022 – Dec 2022

Boston University, Boston, MA

- TA for 'CS 779: Advanced Database Management'. This course covers advanced aspects of database management, query optimization, distributed databases, data warehousing, and big data databases.
- TA for 'CS 677: Data Science with Python'. The course covers Python tools and techniques for data analysis and help build statistical, visualization and data science skills for effective use in a variety of applications.
- TA for 'CS 521: Information Structures with Python'. This course covers the concepts of the object-oriented approach to software design and development using Python.

Manager – Data Science Division

May 2018 - Jul 2021

ICICI Bank Ltd, Bangalore

- Utilized ML to monitor purchase history, credit card transactions to trigger against any fraudulent transactions.
- Build ML models on financial data and analyze financial ratios to decision on credit worthiness of clients
- Managed existing portfolio and added new business portfolio of 2000 Million INR during my tenure.
- Maintained MIS reports, Dashboards and assigned targets and milestones to team members.
- Formulated and analyzed data reports of withdrawal and credit summation of customers using Tableau.

Software Engineering Analyst

Jul 2014 - Jul 2016

Accenture Services Private Ltd, Bangalore

- Created User Stories, Use Cases, FRS and FSD, Test Cases, Test Scenarios, Test Steps.
- Handled an application alone under the project and brought down legacy issues (nearly 15 years old) from 13 to 2 and saved recurring losses of nearly 100 million INR yearly.
- Performed checks on the data, involving Extraction, Manipulation and putting Data Constraints using SQL.
- Worked on Enhancement requirement using Python, and SQL and pushed the changes to production and performed Unit testing and support in QA phase for defects using SQL and Python
- Monitored the database, presented the data to clients using pivot tables and dashboards using Tableau.

SKILLS

Languages/Tools: *Python, SQL, R Programming, Base SAS, DBMS, MS Excel, JIRA, Apache Spark, Hadoop, AWS Redshift, AWS S3, MapReduce, AWS Lambda, GCP*

Data Visualization: *Tableau, Power BI*

Techniques: *Machine Learning, Data Analysis, Data Science, Data Engineering, Data Mining, Data Wrangling, Forecasting, Time Series Analysis, Natural Language Processing, Clustering, Web Analytics, Agile, Scrum*

RESEARCH & PUBLICATIONS

- Using machine learning for healthcare treatment planning Apr 2023
- Lac-Phe mediates the effects of metformin on food intake and body weight Apr 2024
- Protein biomarkers of adverse clinical features and events in sarcomeric hypertrophic cardiomyopathy Dec 2024
- Lac-Phe mediates the anti-obesity effect of metformin Nov 2023
- Circulating metabolite biomarkers of glycemic control in youth-onset type 2 diabetes Sep 2024
- Utilization rates of SGLT2 inhibitors and GLP-1 receptor agonists remain significantly lower than those for other cardioprotective medications: Results from a retrospective Nov 2024
- Acute cold exposure in humans shifts the circulating proteome to a cardioprotective and anti-aging profile Jul 2025
- Plasma Proteomics Reveals Dysregulated Pathways Across the Spectrum LMNA Cardiomyopathy Jun 2025
- SECTM1 Regulates Monocyte Levels and Is Associated With Incident Coronary Heart Disease Sep 2025
- Surface electrical impedance myography detects disease in an adult-onset SOD1-G93A zebrafish model of amyotrophic lateral sclerosis Oct 2025
- Performing Compound Motor Action Potential Measurement in Zebrafish: A Description of Methodology and a Comparison Between Healthy and ALS-Affected Animals Aug 2025
- Longitudinal protein associations with impaired fasting glucose (IFG) and diabetes in MESA Nov 2025
- N-Palmitoyl Glutamine Is a Candidate Mediator of Cardiorespiratory Fitness Nov 2025

FUNDING

- Independently awarded analytic funding for National Institute of Health (NIH) MoTrPAC Study of human exercise.

PROJECTS

- Data Warehouse Creation and visualization using Python and SQL
- Automatic Detection of Diabetic Retinopathy using CNN and ANN using Python
- Netflix Recommendation using AWS S3, Redshift, AWS Lambda, AWS Glue, AWS Sage maker
- Real-Time Detection of American Sign Language using Advanced ML in Python
- Yellow Cab Fare Prediction through R and various machine learning models
- Pose Estimation and Clustering similar exercises using Python

LICENSES & CERTIFICATIONS

- Lean Six Sigma Green Belt Executive HENRY HARVIN EDUCATION
- SAS Certified Base Programmer for SAS 9.0 SAS GLOBAL CERTIFICATION
- Google Data Analytics Professional Certificate GOOGLE, Coursera